



Introduction to Python



Icebreaker: *One highlight* of their summer vacation?





Mini-Lesson – Before the Warm-Up

- User Input – We are able to ask the user for input or ask the user for information.

input() – Getting Information from the User

```
username = input("Enter username:")
```

- Strings- We are able to text stored in variables.

```
color = "Blue"
```

```
name = 'Alex'
```

Strings can be combined or display with + or f-strings:

```
print("Hello " + name + "!")
```

```
print(f"Hello {name}!")
```

If:

- **A statement that lets your program make a choice.**
- It checks a condition, and if the condition is **True**, it runs the code inside.

Else:

- A statement that tells your program **what to do when the if condition is False**.
- Always comes **after an if**.

Snack Time

```
# Ask the user if they want chips
```

```
snack = input("Do you want chips? ")
```

```
# Check the user's answer
```

```
if snack.lower() == "yes": # If the answer is "yes"
```

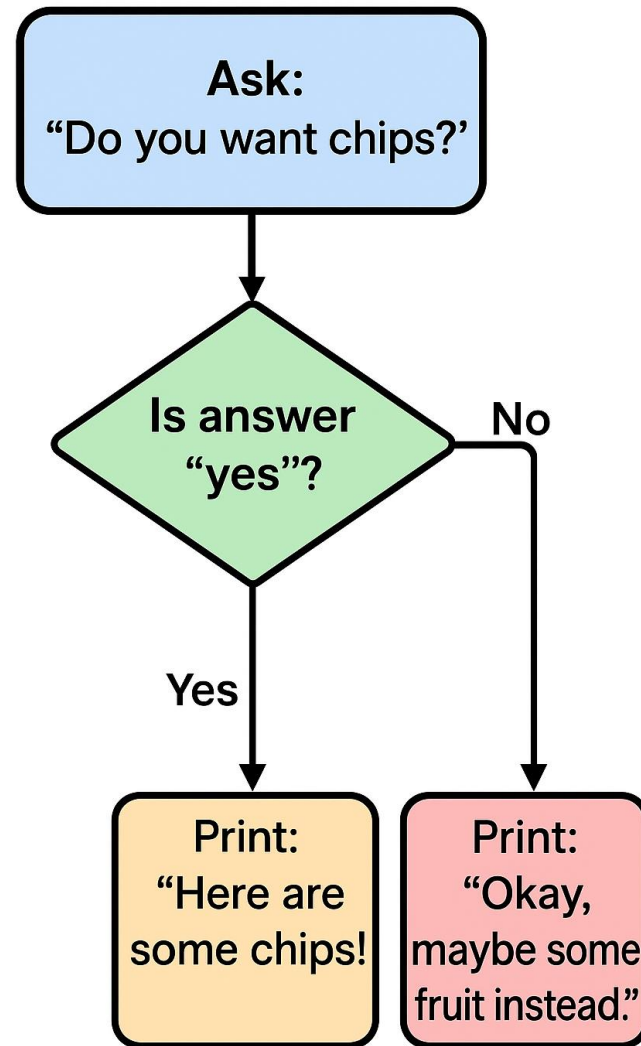
```
    print("Here are some chips!") # Do this if condition is True
```

```
else: # If the answer is not "yes"
```

```
    print("Okay, maybe some fruit instead.")# Do this if condition is False
```

Flow chart

A flow chart is a visual way to show steps in a process or how something works using shapes and arrows.



Process / Action

- **Shape:** Rectangle
- **Purpose:** Represents a step or action in the process.

Decision

- **Shape:** Diamond
- **Purpose:** Represents a choice or question that can lead to different outcomes.

Arrow / Connector

- **Purpose:** Shows the direction of flow from one step to another.

Warm-Up Question

```
name = input("What is your name? ")
color = input("What is your favorite color? ")
if color.lower() == "blue":
    print(f"Hey {name}, you're a cool person!")
else:
    print(f"{name}, {color} is a unique choice!")
```

- What happens if someone types **"Blue"** vs **"blue"**?
- Why do we use `color.lower()`?
- What does `f"{name}"` do in the print statement?

Instructions

- **Form Groups:** Please get into your groups.
- **Start the Competition:** Once your group is ready, we will begin the activity.
- **Complete the Task:** Each group will work to complete the given task correctly.
- **Winner Selection:** The group that completes the task first and correctly will get the opportunity to choose a topic of interest for the next activity. (**looks into datasets** and analyze the information.)

Topic Exploration: The chosen topic will be explored as a class. Possible topics include:

- Movies
- Sports
- Music charts
- Video games

Recognition: The winning group will have their name displayed as the winners of this activity.

Name & Age Bot

- Asks the user for their **name**.
- Asks the user for their **age**.
- Prints a greeting message using the name and age provided.

Math Practice App

- **Write the Program:**
- Ask the user a math question (e.g., $5 + 3$).
- Check if their answer is correct using an `if` statement or a `while` loop.
- If the answer is wrong, let them try again until they get it right.
- Display a congratulatory message when the answer is correct.
- **Test Your Program:** Run it to make sure it loops correctly and responds when the user answers correctly.
- **Optional Extension:**
- Add more questions.
- Track the number of attempts.
- Give hints if the user keeps answering incorrectly.

Exercise 1: Calculate the Average of a Fixed List

```
def get_average(scores):  
    total = sum(scores)  
    return total/ len(scores)  
  
grades = [49, 94,68,42]
```

Task:

Use the given list of grades `[85, 90, 78, 92]` and write a function `get_average()` that calculates the average score. Print the result.

Exercise 2: Ask the User to Enter Scores and Calculate the Average

Write a program that asks the user to input 4 test scores (use a for loop). Store the scores in a list called `user_scores`. Use your `get_average` function to calculate and print the average.

- Hint: Use `int(input())` inside the loop.
- Bonus: Ask the user how many test scores they would like to enter. Then, collect those many scores and calculate the average.

Hint:

- Use `input()` to ask the user for the number of scores and convert it to an integer with `int()`.
- Use a for loop that runs as many times as the number the user enters.
- Inside the loop, ask the user to enter each score and add it to a list.
- Finally, call the `get_average()` function with your list and print the result.